

Credit Portfolio Risk Analysis

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Portfolios: Portfolio 1&2

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Contents

1	Introduction	2
2	Data	2
3	Methodology	3
4	Results and Discussion	5
5	Conclusion	6

1. Introduction

Credit risk is the risk of loss arising from an obligor's failure to meet its financial obligations. At the portfolio level, this risk depends not only on individual default probabilities, but also on the dependence among obligors, since common systematic shocks may lead to clustered defaults and severe tail losses. Therefore, this report evaluates both expected loss and tail risk.

This report analyses the one-year credit risk of Portfolio 1 and Portfolio 2 using a one-factor Gaussian threshold model. The raw datasets are cleaned and integrated to construct the key model inputs, including probability of default, exposure at default, loss given default, portfolio weights, and the dependence parameter. The selected model is appropriate because it represents default as a threshold-crossing event and introduces correlated defaults through a common systematic factor, allowing portfolio tail losses to be simulated.

The simulated loss distributions are used to compute expected loss, Value at Risk, and Expected Shortfall. Expected loss measures the average portfolio loss, while VaR and ES summarise the upper tail of the loss distribution. By comparing these measures, the report identifies which portfolio has higher credit risk and explains the economic intuition behind the difference.

2. Data

This analysis uses five datasets to construct the modelling inputs for Portfolio1 and Portfolio2. The portfolio files provide obligor-level exposure information, including entity codes, exposure at default, booking dates, and internal comments. The entity information dataset contains firm-level characteristics, including industry, region, size bucket, loss given default, and base probability of default information. Stock return and industry index return data are used to estimate the dependence parameter, while the historical default dataset provides default event records for supporting probability of default estimation. These datasets are integrated to construct the key inputs for credit portfolio simulation: EAD, PD, LGD, portfolio weights, and the dependence parameter.

The raw data contain several inconsistencies that need to be addressed before modelling. First, entity codes appear in inconsistent formats, including lower-case letters, extra spaces, and missing hyphens. Entity codes are standardized by removing unnecessary spaces, converting all letters to upper case, and correcting missing hyphens where necessary. Second, industry labels appear in multiple formats, such as abbreviated labels, labels without spaces, and different capitalizations. These are standardized into consistent categories because industry information is later used both for LGD imputation and for matching firms with industry return

series. Third, date variables appear in mixed formats across the return and default history datasets, so they are converted into a consistent datetime format. Finally, duplicated rows are removed to avoid double-counting observations during parameter estimation and simulation.

After preliminary cleaning, each portfolio is merged with the cleaned entity information dataset using `entity_code`. This creates a modelling table in which each obligor is linked to its exposure, industry, size bucket, LGD, and PD information. The missing-value treatment relies on several economically motivated assumptions rather than purely mechanical imputation rules. Missing EAD values are imputed using the average EAD of obligors in the same size bucket, based on the assumption that exposure size is more closely related to firm size. Missing LGD values are imputed using the average LGD within the same industry, reflecting the assumption that firms in the same sector may have similar recovery characteristics and loss severity. Missing PD values are first supplemented using historical default frequencies estimated from the default history dataset. If the resulting PD is zero, a small positive floor is applied to avoid numerical problems in the inverse-normal threshold calculation used in the simulation model.

The dependence parameter is estimated using market return information. Firm-level stock returns are matched with the corresponding industry index returns by date and industry. The absolute correlation between the firm return and the industry return is used as a proxy for the obligor's exposure to systematic industry risk. Missing return observations are not directly imputed; instead, correlations are estimated using available matched return pairs after removing missing observations. This avoids introducing artificial return values while still allowing observed market co-movement to inform the dependence structure.

After these procedures, portfolio weights are then calculated as each obligor's EAD divided by total portfolio EAD. These cleaned and integrated modelling tables provide the direct inputs for the one-factor Gaussian threshold model described in the next section.

3. Methodology

This report uses a one-factor Gaussian threshold model to simulate the one-year credit loss distributions of Portfolio 1 and Portfolio 2. The one-factor framework is appropriate for portfolio credit risk because it introduces dependence among obligors through a common systematic factor. This structure is closely related to the Vasicek-style credit portfolio model, which provides a tractable framework for modelling correlated defaults and portfolio loss distributions (Vasicek, 2002)^[1]. It is also connected to credit VaR and capital modelling

frameworks discussed by Gordy (2003)^[2]. VaR and Expected Shortfall are used as standard tail risk measures in quantitative risk management (McNeil, Frey, & Embrechts, 2015)^[3].

For each obligor i , default is represented by a binary variable Y_i , where $Y_i = 1$ if the obligor defaults and $Y_i = 0$ otherwise. The marginal probability of default is denoted by p_i . A latent credit variable X_i is introduced to represent the underlying credit condition of the obligor. Default occurs when this variable falls below an obligor-specific threshold:

$$Y_i = \mathbf{1}(X_i \leq d_i).$$

Assuming that X_i follows a standard normal distribution, the threshold is determined by the inverse standard normal cumulative distribution function: $d_i = \Phi^{-1}(p_i)$. This transformation ensures that the simulated marginal default probability is consistent with the obligor-specific PD. Dependence between obligors is modelled through a common systematic factor:

$X_i = \rho_i F + \sqrt{1 - \rho_i^2} \epsilon_i$, where $F \sim N(0,1)$ represents the common systematic factor and $\epsilon_i \sim N(0,1)$ represents the idiosyncratic shock of obligor i . The parameter ρ_i measures the sensitivity of obligor i to the systematic factor. A higher ρ_i implies stronger exposure to common risk conditions and a greater possibility of clustered defaults.

The required model inputs are EAD_i , LGD_i , p_i , and ρ_i . EAD is obtained from the portfolio files, while LGD and base PD are obtained from the entity information dataset. When base PD is missing, the historical default frequency is used as an empirical proxy:

$$\hat{p}_i = \frac{\text{Number of observed defaults for obligor } i}{\text{Number of valid historical observations for obligor } i}.$$

If the estimated PD equals zero, a small positive floor is applied to avoid numerical problems in the inverse-normal transformation. The dependence parameter ρ_i is estimated from market return data. Firm-level stock returns are matched with the corresponding industry index returns, and the absolute correlation between the two return series is used as a proxy for exposure to systematic industry risk: $\hat{\rho}_i = |Corr(R_{i,t}, R_{\text{industry},t})|$.

After the parameters are constructed, Monte Carlo simulation is used to generate the one-year loss distribution. In each simulation trial, the common factor and idiosyncratic shocks are simulated, default indicators are determined by the threshold rule, and obligor-level losses are calculated as: $Loss_i = EAD_i \times LGD_i \times Y_i$.

The total portfolio loss is: $L = \sum_{i=1}^m EAD_i \times LGD_i \times Y_i$.

To compare portfolios with different total exposures, the loss rate is also calculated: $L_{\text{rate}} = \frac{L}{\sum_{i=1}^m EAD_i}$.

The analytical expected loss is computed as: $EL = \sum_{i=1}^m EAD_i \times LGD_i \times p_i$.

In rate form, this is: $EL_{\text{rate}} = \sum_{i=1}^m w_i \times LGD_i \times p_i, w_i = \frac{EAD_i}{\sum_{j=1}^m EAD_j}$.

Expected loss describes the average portfolio loss, but it does not measure the severity of adverse scenarios. Therefore, Value at Risk and Expected Shortfall are computed from the simulated loss distribution. The Value at Risk at confidence level α is defined as: $VaR_{\alpha}(L) = F_L^{-1}(\alpha)$, and Expected Shortfall is calculated as: $ES_{\alpha}(L) = E[L \mid L \geq VaR_{\alpha}(L)]$.

In this analysis, 95% and 99% VaR and ES are calculated for both portfolios. These measures allow the comparison of average loss and tail risk. This distinction is important because the common systematic factor can generate correlated defaults, leading to losses much larger than expected loss alone would suggest.

4. Results and Discussion

Table 1 summarises the main simulation results for Portfolio 1 and Portfolio 2. Two portfolios contain 30 obligors, but Portfolio 2 has a larger total exposure and a higher overall risk level. Portfolio 1 has a total EAD of approximately 680.47 million, while Portfolio 2 has a total EAD of approximately 766.53 million. After adjusting for portfolio size, Portfolio 2 remains riskier: its simulated expected loss rate is 1.80%, compared with 0.90% for Portfolio 1. The simulated expected losses are also close to the analytical expected losses, suggesting that the Monte Carlo simulation is consistent with the PD, LGD and EAD inputs used in the model.

Table 1. Comparison of Portfolio Risk Measures

Portfolio	Simulated EL Rate	95% VaR Rate	99% VaR Rate	95% ES Rate	99% ES Rate
Portfolio 1	0.90%	4.39%	7.94%	6.62%	10.37%
Portfolio 2	1.80%	8.19%	15.28%	12.55%	19.70%

The results show that Portfolio 2 is riskier across both average and tail-risk measures. For Portfolio 1, the 99% VaR rate is **7.94%** and the 99% ES rate is **10.37%**, indicating that losses can become much larger than the expected loss under adverse scenarios. For Portfolio 2, the corresponding values increase to **15.28%** and **19.70%**, respectively. This means that Portfolio 2 not only has a higher average loss, but also a much heavier loss tail.

The difference is also reflected in the simulated number of defaults. Portfolio 1 has a mean number of defaults of approximately **0.58**, while Portfolio 2 has a higher mean number of defaults of approximately **1.10**. This suggests that Portfolio 2 is more vulnerable to default clustering. Under the one-factor Gaussian threshold model, such clustering is driven by the common systematic factor: when the common factor deteriorates, several obligors may default together, leading to large portfolio losses.

From a risk-management perspective, Portfolio 2 would require a larger capital buffer than Portfolio 1. Expected loss is useful for pricing, provisions and reserves, but it does not fully capture unexpected losses in adverse scenarios. VaR and ES are more informative for economic capital planning because they summarise the upper tail of the loss distribution. In particular, the higher 99% ES rate of Portfolio 2 indicates that once losses exceed the 99% VaR threshold, the average tail loss is substantially more severe.

Overall, the results suggest that Portfolio 2 has a less favourable combination of default probabilities, loss severities, exposure distribution and systematic dependence. Therefore, Portfolio 2 should be regarded as the riskier portfolio, especially from a tail-risk and capital-planning perspective.

5. Conclusion

This report examined the one-year credit risk of Portfolio 1 and Portfolio 2 using a one-factor Gaussian threshold model. After cleaning and integrating the raw datasets, the key inputs, including EAD, PD, LGD, portfolio weights and the dependence parameter, were constructed for simulation. The results show that Portfolio 2 is riskier than Portfolio 1. Its expected loss rate, VaR and Expected Shortfall are all substantially higher, especially at the 99% confidence level. This indicates stronger tail risk and greater exposure to default clustering. Overall, Portfolio 2 would require a larger capital buffer for risk management.

References

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